Outsourcing and Time-related Dimensions of Job Quality: Evidence from the Facility Management Business

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1. Introduction

The decentralization and outsourcing of economic activities are not only a defining feature of the network society, but also has experienced an important expansion in its mechanisms, the actors engaged and its results (Harrison and Kelley 1993; Autor 2003). Nowadays, there is an extension of new forms of outsourcing mostly based on the opportunities offered by new technologies, including the growth of (bogus) self-employment, crowd employment through platforms and the expansion of facility management companies (Drahokoupil 2015; Aloisi and De Stefano 2019; Beuker et al. 2019; Molina et al. 2023). It is increasingly common to find companies relying on several of these mechanisms. Moreover, it is not rare to observe provider organizations further outsourcing some of the services they provide to clients, therefore contributing to a cascading subcontracting effect. The changing and varied nature of outsourcing practices is explained by the companies' features, the sectoral background, the implementation of technologies facilitating remote and automated control, and the institutional and regulatory framework. Moreover, outsourcing drivers are equally varied, relaying on gaining flexibility and efficiency, while reducing costs and evading fiscal or labour market regulations (Doellgast and Gospel 2013; Godino and Molina 2022).

In this context, the growth and expansion of facility services companies is probably one of the most interesting developments in relation to outsourcing and its implications for employment. Outsourcing is far from being a new phenomenon, but only since the early 2000s has there been a significant expansion of facility management companies in Europe, though uneven across countries. Facility management (FM) companies provide integrated services and the management of those services to other companies or facilities. Even though the integration and management components constitute the distinctive traits of this form of outsourcing, the reality is that FM firms very often compete as any other service company providing just one service to client companies. As a matter of fact, rather than competing in the integrated service providers market, FM companies often take advantage of the diversity of services provided to engage in cost competition, thus undermining the working conditions of some groups of workers through the application of lower standards in collective bargaining. While most of the literature on industrial relations has focused on these gaps between outsourced and internal workers in relation to

collective agreement coverage (Godino and Molina 2022) and wages (Dube and Kaplan 2010), few have addressed an increasingly key aspect of job quality: the time dimensions of employment (Recio et al.2015).

This chapter examine firstly the degree of outsourced workers in selected occupations common in facility services. Secondly, it observes working conditions of facility management workers in Spain in comparison to in-house workers to explore gaps in relation to working time, stability, and work-shifts. The analysis serves to illustrate the differential conditions in these three dimensions, although showing better standards in comparison to in-house workers for certain dimensions (e.g. permanent work) while observing worse job quality in relation to the extent of part-timers, having a later impact on salaries.

The first section of the chapter discusses FM in light of the literature on the characteristics and impact of outsourcing and the specific characteristics of FM. Section two then discusses the methodology and the analytical strategy used, including the job quality approach adopted when studying time dimensions. Section three shows the results of the logistic regression models carried out. This is then followed by some concluding remarks, showing three different scenarios for outsourced workers in FM businesses in relation to time-related dimensions of job quality.

2. Outsourcing practices and the facility management business

Most production and service provision processes are sequential and require establishing the so-called value chain, i.e., a relationship between the different processes required to produce a good or service (Huws 2012; Antràs and Chor 2013). The design of the economic value-added formula in the 1990s advocated that companies were more efficient when they removed as many items as possible from their balance sheets (Marchinton et al., 2005). Outsourcing should accordingly be interpreted first and foremost as a business strategy with a financial/economic rationale. First, outsourcing is aimed at reducing costs in the short term (suppliers base their business on economies of scale, reusing models already developed for different clients). Second, it is also aimed at transforming companies' long-term cost structures by eliminating the fixed structures that support the internal control of the outsourced processes. Third, outsourcing constitutes a mechanism to enhance organizational flexibility, as it allows for a quicker adaptation to production requirements. Finally, the greater specialization of provider companies allows for efficiency gains (Kotlarsky et al. 2011; Doellgast and Gospel 2013; Godino and Molina 2022).

Since the 1980s, the modularization of tasks and processes in the value chain has been required to eliminate workers engaged in ancillary activities through outsourcing practices (Harrison and Kelley 1993; Flecker and Meil 2010). In turn, this allowed the organization of production around

networks of supplier companies, increasingly favouring the market over organizations' internal competences (Ricart et al. 2011; Kirchner 2015). Through outsourcing, the delivery of non-core activities is extracted from the central organizations as well as the responsibility over labour and social issues of the workers completing these activities, thus dualizing workers' organizational and employment relations (Rivero et al., 2006; Doerflinger and Pulignano 2015; Mori 2015; Moreno-Colom et al. 2014). In this way, the reduction of labour-related risks and costs (including wages, absenteeism and conflicts, among others) mainly through vertical disintegration strategies is an important driver for outsourcing, which moves workers out of well-organized sectors and/or organizations to spaces with weaker structures of worker representation and lower capacity for collective action (Doellgast et al. 2009; Huws 2012; Recio et al. 2015; Mori 2015; Hyman 2015; Benassi and Dorigatti 2020). Therefore, client companies tend to trade the control of some internal work processes for low-pay strategies (Godino and Molina 2022).

FM businesses appear to be opportunities for companies to compensate for this organizational fragmentation of work processes by integrating most of their outsourced activities in the same supplier firm. Therefore, FM allows simplification of interorganizational coordination and reduction of transaction costs; either FM firms provide the outsourced services, or they procure services from external suppliers and manage the output from the service providers. Moreover, the management and delivery of these integrated outsourced services is increasing with the internationalization of contract procurement by multinationals, as FM companies sign contracts with large clients at the European level, facilitating standard processes with providers.

2.1. Facility Management: definition, characteristics, and practices

The conceptual and empirical delimitation of FM poses several challenges, as it is a relatively new business strategy that spans traditional sectoral classifications and points to greater changes in relations between organizations, space, and work (Grimshaw 1999). Moreover, FM companies adopt very different practices and strategies across countries, therefore making it even more difficult to define FM based on a set of common dimensions. Notwithstanding these problems, there is some consensus around the idea of an integrated management (and delivery) of non-core services and processes supporting the core business of an organization (Redlein 2004; Ferri and Pala 2009; Redlein and Zobl 2014; IOS 2017). What is accordingly distinct from other service providers is the effective integration and coordination of these ancillary activities (Avenali, Matteucci and Nonino 2010). However, a relevant trend within FM managerial literature highlights that the coordination, rather than the provision of services, is the "bread and butter" of FM activity and notes its potential in contributing to wider business outcomes (Drion et al. 2012).

In light of the above, a distinction can be made between internal and external facility management. FM activities (i.e., those tasks developed by professionals integrating and coordinating outsourced services) are not completely outsourced in the European market. A total of 49.5% of facility managers develop their activity internally (Global FM 2018). Internal facility managers operate within client organizations while coordinating external supplier companies. In contrast, outsourced facility managers externally coordinate the integrated supply of outsourced services. These services can be delivered by the same company that coordinates externally or by other provider firms contracted by the FM company.

In terms of services provided, facility management traditionally covered infrastructure and maintenance, equipment and furniture management, efficient resource supply, waste removal and security services. However, other functions, such as supervising environmental aspects or managing IT systems, are increasingly common, especially in those countries with more developed FM. This wide range of activities supplied by FM companies explains the growth of FM in recent years and why it is becoming one of the largest business services markets in Europe (Global FM 2016, 2018).

FM companies are generally classified according to their territorial scope (distinguishing companies that supply outsourced services to client organizations at the local, regional, and global levels) but also in relation to the extent and complexity of activities supplied. In that sense, the literature distinguishes between multi-service suppliers, facility services suppliers and FM companies based on the number of activities covered and the level of integration and coordination between the services provided (Ferri and Pala 2009; Hodge et al. 2014).

Moreover, there are important differences across countries in relation to the development of the FM business and the type of services provided, allowing us to differentiate several market stages (Ferri and Pala 2009; Teichmann 2009): pre-emerging markets (mostly composed of single-service supply firms and client-supplier relationships based on multiple single-service and short-term contracts), emerging markets (with an incipient set of multi-service supply companies and a fragmentary supply concerning activities, territory coverage and clients, as well as a small number of multinationals with pan-European contracts), developed markets (where single-service supply co-exists beside multi-service supply, consolidating large supply operators) and advanced markets (with a large amount of integrated-service supply and the implementation of public-private partnership projects).

The services supplied by FM firms are generally labour intensive and carried out by predominantly unskilled workers who are typically paid low wages (Price 2015), and the rates of employment are higher for women and migrant workers. In that sense, a facility management business gathers activities and workers frequently ascribed to the secondary segments of labour

markets within the same organization, thus functioning as a parallel company of the client organizations supplying the ancillary activities. The question is accordingly whether FM contributes to maintaining the segmented pattern in the labour market or facilitates an upgrade in workers' positions (Godino and Molina 2022).

This potential pattern of segmentation is reflected in the labour practices of FM companies in Spain. The country has developed notably its FM market since the late 2000's. This was followed by attempts to bargain a specific FM business collective agreement. Sectoral coordination of collective bargaining prevailed until the 2012 labour law reform opened the door to a decentralization of collective bargaining through the extension of opting out clauses in company-level agreements. This allowed FM companies to prioritize company-level agreements to set lower standards than sectoral agreements. Trade unions responded to these challenges first by appealing to justice. Second, gradually introducing clauses aimed to solve the problem of indeterminacy of the collective agreement to be applied to FM workers while discouraging the arbitrary application of collective agreements in the FM business. Meanwhile, some of the largest FM employers had contacts to launch their own employer organization to promote sectoral negotiations for the FM businesses. According to employers, having a specific sectoral collective agreement would allow all workers in FM companies to enjoy similar conditions. However, trade unions interpreted this move as a further attempt at eroding employment conditions. This proposal was finally dismissed due to unfruitful negotiations with unions and the multiple judgments cancelling FM company agreements (Godino and Molina, 2022).

3. Time dimensions of job quality

To our knowledge, the literature on the association of outsourcing with quality of employment has mainly focused on wages, whereas other labour inequalities between internal and external workers have been hardly studied: for example, identifying a common increasing workload intensification of outsourced public services in different national industrial relations regimes (Mori 2019). Or finding how subcontracting companies in the call centres sector offer less secure jobs, higher use of invasive monitoring practices, and less worker control over schedules and breaks compared to similar 'in-house' call centres (Doellgast and Pannini 2015). This happens particularly with those aspects of job quality inherently related to the organization and quality of time (Moreno-Colom, Recio and Godino, 2015). The debate on working time has always been present since the origin of factories (Hobsbawn 1984), with an initial academic focus on the duration of working time (Boulin 2001), which has subsequently adopted a more multidimensional approach (Muñoz De Bustillo, Fernández-Macías, et al. 2011).

Ancillary activities (e.g. industrial cleaning) demand flexible, customer-driven work in specific ways. As many service activities are outsourced, there is a double pressure on time schedules designed by both the firm and its clients. Employees are usually required to adapt to time-specific arrangements and have limited capacity to bargain these conditions. Each branch of activity has its own model of working time and employees consider it as the sectoral standard, though they sometimes aim to adapt their working hours to their own requirements (Recio et al. 2015).

Among the several social sciences traditions to analyse job quality, the institutional and the work-life balance approaches are particularly key concerning the consideration of time dimensions. These traditions include aspects related to contractual stability, working time duration and scheduling to understand what job quality is and how it might be measured (Muñoz de Bustillo et al. 2011). This has been a path followed by the most important studies addressing the multidimensionality of job quality (Eurofound 2002; Leschke, Watt and Finn 2008; Prieto et al. 2009; Grimshaw and Lehndorff 2010; Muñoz de Bustillo et al. 2011).

Labor market segmentation theories have traditionally considered stability, along with wages, as one of the fundamental dimensions that shape and differentiate each of the segments into which the labour market is divided (López 1994; López et al. 1998). When the job is not temporary, it leads to the creation of communities with internal rules that are different from the rest (Doeringer and Piore 1970). In this way, we understand stability as the different capacity to maintain employment that the various groups of workers have (López, et al. 1998).

Beyond these formal aspects that shape the duration of the employment contract, various factors have substantially modified the working time for decades: the crisis in industrial employment, the rise of the service sector, and the quantitative increase of women wage-earners (Miguélez et al. 2005). Thus, working time has become one of the most discussed aspects of job quality. A dimension under which not only issues related to the number of hours worked per week, but also concerning the regulation and organization of annual working hours, scheduling, and unsocial shifts (Miguélez et al. 2005). The increasing demand of availability of facility services companies might imply challenges for the job quality of these workers.

4. Data and methods

4.1. Data

This chapter was aimed at answering the following research questions: (1) What is the degree of outsourced workers in occupations under FM companies? And (2) what is the relation between outsourcing and time dimensions of job quality in these occupations? To answer these questions, we employed two data sources. Firstly, we used the annual version of the Spanish Labour Force

Survey (EPA) microdata for 2018 and 2020 (INE, 2019; INE, 2021). These cross-sectional datasets pool in the same file the subsample of each of the four quarters that is subjected to the annual questions. Thereby, in the microdata coming from LFS, one fourth of the individuals answers the questions during the first quarter, one fourth during the second, one fourth during the third, and one fourth during the fourth. Secondly, we also exploited the crosstabulations of the Population and Housing Censuses of Spain for 2011 and for 2021 (INE, 2012; INE, 2023). We could not use their microdata to run the regressions, since 3-digits versions of the occupational or economic activity classifications were not available. Nevertheless, we used their published tables to assess the reliability of the survey estimates. Furthermore, the Census of 2021 has the advantage that it collected the occupational and sectoral information from administrative registers and not from personal surveys. This avoids measurement errors that might occur with workers confusing occupational classification and/or the specific sectoral classification of their employers.

Following the strategy of Dube and Kaplan (2010), we defined an outsourced worker as the one who "works for an employer that provides labour services mainly as an intermediate input to a primary firm, when that individual could in principle provide the same labour services as a direct employee of the primary firm" (p. 290). First, this analysis crosscuts the economic sector of the employer and the occupation of the employee. Then, it qualifies as outsourced any worker employed in one of these sectors who is exclusively engaged in the provision of labour-based services. In our case, the sector is measured through the CNAE-09 at 3 digits, the Spanish classification of economic activities, which is a further disaggregation of NACE Rev. 2. The occupation of workers is measured through the CNO-11 at 3 digits, the Spanish classification of occupations, which presents a close correspondence to the ISCO-08.

Specifically, we focused on three occupations within facility services because they allowed a clear identification of outsourced workers. Cleaners, with the CNO-11 equal to 921, are considered as outsourced if they worked in the economic activity 812. Security guards, those with CNO-11 equal to 594, are described as outsourced if the sector of their firm was 801, 802, or 803. Gardeners, with CNO-11 equal to 612, are categorized as outsourced if they were employed in a company with sector 813.

We considered other variables during the analysis as well. Regarding the socio-demographic variables, some of them were recoded due to sample size following common classifications like ISCED for levels of education or NUTS for regions. To analyse some time dimensions of job quality, we followed the work of Muñoz de Bustillo et al. (2011). Specifically, we studied three dimensions: contractual stability, working time duration, and working time scheduling. We retained the subsample of employed individuals and then we constructed the first two outcomes through binary variables that registered if the employee had a temporary contract (first dimension)

and was part-timer (second dimension). The third outcome was operationalized through a *good schedule* index following Muñoz de Bustillo et al. (2011). This variable was created as the sum of the scores of four items: working at night, in the evening, on Sundays, and on Saturdays. The person received for each item 1 point if she/he declared that never did, 0.5 points if he/she said that did it once a month, or 0 points when she/he did at least twice a month.

4.2. Analytical strategy

The first analysis estimated the proportion of outsourcing in three facility services occupations. We considered the sampling design of the surveys, so estimators of the Horvitz-Thompson class were used to take into account the non-equal sampling weights of each individual (Lumley, 2010). After characterizing the temporal evolution of the prevalence of outsourcing by occupation, we crossed the outsourced status with other sociodemographic variables through contingency tables. Row proportions were estimated, and inference was carried out with the Rao-Scott second-order correction to the Pearson chi-squared test (Scott, 2007).

The second analysis explored the association between being outsourced and the abovementioned dimensions of job quality. Our goal was to estimate the predictive ability of outsourcing, so no causal model was required[1]. To quantify the practical significance of outsourcing as a predictor versus other socioeconomic variables, we performed a relative importance analysis. This analysis considers not only the predictive ability a covariate may have in companion of other predictors (in a multivariable model), but also on its own, as a good proxy of other predictors of the outcome (Johnson & LeBreton, 2004). Due to the survey character of our main dataset, we have opted for combining the results coming from multivariable and bivariable analyses instead of using other more novel interpretability statistic.

Firstly, for the partial associations, logistic regression models were estimated for the binary outcomes and a linear regression model for the good schedule dimension. The simplicity of the model structure was dictated by the nominal scale of most of the candidate predictors and by the reduced sample size we have. The specification followed the guidelines of Harrell (2015) and Riley et al. (2019a; 2019b) for multivariate prediction models by determining in advance the number of degrees of freedom we may spend. To facilitate the interpretation of coefficients, all estimates have been expressed in average marginal changes[2]. The standard error estimation considered the sampling weights thanks to the {survey} package implemented by Lumley (2010).

[1] Strictly speaking, we implicitly assumed a specific causal model for the missing data: missing completely at random (Mohan & Pearl, 2021). Still, the objective of the chapter was not effect estimation.
[2] These quantities are more known as average marginal effects or average partial effects. We have preferred to substitute the last term by *change* to highlight the non-causal goal of our research.

Secondly, for the total associations, we have estimated Cramér's V for each predictor and each time dimension of job quality. This statistic has been adapted to the complex sampling setting following the recommendations of Lumley (2022). The advantages of this quantity are that it is easily understandable and that it is simple to adapt to predictors with interval scale.

5. Results

In this section, we start by providing estimates regarding the extent of outsourcing for each job. Following that, we share findings from a multivariable model that measures the predictive power of certain covariates on time-related aspects of job quality. We specially focus on the importance of the outsourcing status versus other predictors easier to measure.

5.1. The prevalence of outsourcing

The first objective of this research was to estimate the proportion of outsourcing in the three occupations of interest. Table 1 shows the percentage of outsourced workers for each job estimated with the Labour Force Survey (2018 and 2020) and with the Census (2011 and 2021). At least from 2018, for cleaners and security guards, it is more common to be employed by a provider company than to be an internal worker. One out of two cleaners served as outsourced worker in Spain during 2018. Comparing the survey and the census proportions, the estimates are quite robust to the source and the year of measurement from 2018 onwards. These findings are consistent with the literature, identifying a trend to outsource ancillary services to businesses (López 1994; Ahmed et al. 2011; Kirchner, 2015) particularly being the case of cleaners and security guards (Tregenna 2008).

Table 1 – Proportion of outsourced workers by occupation (3-digits)

	Census 2011	*LFS 2018*	*LFS 2020*	*Census 2021*
Cleaners	21.24	52.32	57.88	52.91
		(1.75)	(1.81)	
Security guards	28.53	74.39	84.92	
		(3.5)	(3.01)	
Gardeners	27.66	37.21	37.80	38.92
		(4.25)	(4.31)	

Note: Own elaboration based on INE (2012), INE (2019), INE (2021), and INE (2023). The cells of the columns that employed survey data show the estimated proportion with its standard error in parenthesis.

Looking at security guards, three out of four workers were outsourced according to the last estimate before the covid crisis. This proportion of outsourcing is the one experiencing the largest change from 2011 to the last estimate in 2020. It is also the occupation with the highest prevalence of outsourcing between the three facility services occupations analysed. On the contrary, outsourcing is not majoritarian in the case of gardeners. The last estimate indicates that around 39 % of the persons that had this occupation were outsourced. If we focus on the temporal evolution of this proportion, there is a smaller change between 2011 and 2021 compared with the other occupations. It is also interesting to observe how this proportion was not modified once the pandemic impacted in 2020.

Table 2 – Contingency tables for outsourcing and different covariates with LFS-2018

		No	Yes
Sex	Man	42.81	57.19
$\chi^2_{RS2} = 3.685, p = 0.055$	Woman	49.05	50.95
Nationality	EU, except Spanish	47.49	52.51
$\chi^2_{RS2} = 0.110, p = 0.895$	Foreign	48.8	51.2
	Spanish	46.54	53.46
Level-orientation of education	Primary or less	58.64	41.36
$\chi^2_{RS2} = 2.938, p = 0.013$	Lower secondary	48.55	51.45
	General upper secondary	40.45	59.55
	Vocational upper secondary	39.18	60.82
	General tertiary	45.66	54.34
	Vocational tertiary	38.82	61.18
NUTS-1 region	Community of Madrid	35.32	64.68
$\chi^2_{RS2} = 6.997, p < 0.001$	Canarias	59.46	40.54
	Centre	49.03	50.97
	East	43.04	56.96
	Northeast	36.79	63.21
	Northwest	39.62	60.38
	South	59.7	40.3

Note: Own elaboration based on INE (2019). Each cell shows the row proportion in percentage.

Table 2 explores the association between being an outsourced worker and certain sociodemographic variables for the pooled sample of the LFS-2018. Looking at different prevalences by sex, we see that men are more affected by outsourcing, whereas women approximately present equal shares of in-house and outsourced workers. According to the chi-square test, this dependence is statistically significant at the 90 % level. On the contrary, the test cannot reject the independence between being outsourced and the nationality of the individual. The row proportions show that the three categories present more outsourcing than in-house workers, but in the three cases the differences between proportions are quite small.

The next two panels investigate the relationship between outsourcing and two other variables: the level-orientation of education and the big region in which the individual lives. Concerning education, attending to the point estimates of these bivariate associations, the pattern that emerges is quite interesting. Once we disaggregate the levels of education by orientation (vocational or general), there is no linear relationship between the level and the proportion of outsourcing. Those with primary education or less have the lowest prevalence of outsourcing. This proportion keeps increasing in function of the level until it decreases again with those with general tertiary education. Still, the departure of both variables from the independence situation is enough to reject it at the 95 % level through the chi-square test. This would raise the hypothesis that those workers with higher educational attainment may consider certain outsourced and low-skilled workplaces only as in-transition jobs. Therefore, without the possibility of developing careers within these sectors that would allow them to in-house work. Further longitudinal analysis would be required to contract this possibility.

In the last panel, the region of residence is crossed with the outsourcing status. We see how the outsourcing proportion reaches its minimum in the South region (comprising Andalusia, Region of Murcia, Ceuta, and Melilla), while has its maximum in the Region of Madrid. The prevalence is also comparatively high in the Northeast, the region that includes the Basque Country, Navarre, La Rioja, and Aragon. These results are consistent with the stages of FM markets, existing an association between more advanced markets for FM businesses and the amount of wealth created by an economic system (influencing the ability of its firms to efficiently produce high value goods and services) (Ferri and Pala 2009).

5.2. Outsourcing and time-related dimensions of job quality

The second objective of the research was to explore the association between outsourcing and time-related dimensions of job quality. To assess the predictive ability of outsourcing, we have inserted in a multivariable model other features that are usually collected through labour force surveys. We have estimated one model for each outcome variable following the modelling

strategy detailed in the methodology section. To increase the available degrees of freedom, we have pooled the subsample of security guards and gardeners, since the number of events of interest they present were scarce. Note that the different sample sizes available for cleaners and for guards plus gardeners mean that the statistical tests have different power in each case. Statistical significance is more difficult to attain with small sample sizes, so predictors that were significant with the second subsample represent heavier evidence.

Firstly, we focus on the subsample of cleaners. Table 3 shows the results from three models, one for each studied dimension of job quality. These coefficients show the partial association of its covariate with the outcome, that is, having discounted the dependence between predictors. The model in the first column tries to predict the event of temporary employment. The coefficient of outsourcing is statistically significant and presents a negative association. Thereby, in average, being outsourced is linked to 7.2 percentual points less of probability of being temporary worker. Nonetheless, to value the size of the relation, its predictive ability must be compared to other covariates. Holding constant the rest of covariates, the parameter of vocational tertiary education has the largest size. Compared with the primary education, in average, when the individual has such level has 16 p. p. extra of probability of being temporary. There are also two coefficients that are statistically significant showing a negative relationship with the probability of having a temporary employment, the factors of *age* and *Spanish* nationality. As an example, for an interquartile contrast (from 45 to 54 years), the linear parameter of *age* predicts, in average, 9 p. p. less of probability of being temporary. These findings are coherent with the existence of European legislation on employees' rights in situations of changing employers since 1977 under the well-known "Acquired Rights Directive." Since then, mechanisms of social dialogue have further developed in this area, with particular emphasis on the "Transfers of Undertakings Directive" (Council of the European Union, 2001). Subsequently, these directives were implemented in the different member countries and later updated in the Transfer of Undertakings (Protection of Employment) Regulations (Acas, 2024). These directives primarily aimed to ensure not only the continuity of existing contracts before a succession of companies but also the conditions of those contracts. Each Member State was responsible for operationalizing these provisions within their respective legal frameworks. That is the case of Spain with the labour subrogation clauses included in collective agreements, particularly in those sectoral agreements covering outsourced activities (e.g. Industrial cleaning).

Table 3 – Model estimates for cleaners using LFS-2018

	Temporary	Part-time	Good schedule

Outsourced (Yes vs. No)	-0.072** (0.029)	0.137*** (0.034)	0.525*** (0.068)
Sex (Woman vs. Man)	0.017 (0.040)	0.244*** (0.046)	0.036 (0.115)
Nationality (ref.: EU, except Spanish)			
Foreign	-0.046 (0.102)	-0.037 (0.103)	0.360* (0.217)
Spanish	-0.192** (0.096)	-0.037 (0.094)	0.338 (0.211)
Age	-0.010*** (0.001)	-0.006*** (0.002)	0.005 (0.004)
Level of education (ref.: Primary or less)			
Lower secondary	0.044 (0.047)	-0.082 (0.055)	-0.099 (0.103)
General upper secondary	0.030 (0.060)	-0.050 (0.077)	-0.045 (0.146)
Vocational upper secondary	0.039 (0.064)	-0.120 (0.074)	-0.061 (0.135)
General tertiary	0.084 (0.093)	-0.034 (0.118)	0.189 (0.195)
Vocational tertiary	0.159** (0.072)	-0.100 (0.081)	0.021 (0.146)
Num. Obs.	1243	1272	1227
Design-based R^2	0.182	0.094	0.097

Note: Own elaboration based on INE (2019). NUTS-1 region included as control variable. Each cell reports the average marginal change with its standard error in parenthesis. For logistic regression models, the design-based version of the pseudo R^2 of Nagelkerke is shown. * p < 0.1, ** p < 0.05, *** p < 0.01.

The next column presents the estimates for the model that considers *part-time* as the outcome variable. The coefficient of outsourcing is statistically significant at the 99 % level, which indicates that this variable incorporates new information to the model. In average, being outsourced is related to an extra increase of 13.7 p. p. of probability of having a part-time job. This positive association between outsourcing and part-time employment is consistent with several authors remarking how subcontractors employ a larger proportion of part-timers across diverse national and regulatory settings (Shire et al. 2009; Doellgast and Pannini 2015), commonly a cost-cutting strategy by provider businesses (Godino and Molina, 2022). Nevertheless, the variable that incorporates more original information is the *sex*. Table 3 shows that being a woman is related in average to 24.4 p. p. more of probability of being part-time. This result is logical with the largely explored higher impact of part-time work to female-related

activities and occupations (Booth and van Ours, 2013; Recio et al., 2015; Fagan, O'Reilly and Rubery 2020), as is the case of industrial cleaning. Again, the coefficient of age is significant at the 99 % level with a negative association, but this time its point estimate is lower. The rest of coefficients are not statistically significant.

The last column explores the associations for the model that considers the index of *good schedule* as response variable. Note that the outcome variable has interval scale, so coefficients may be interpreted as unitary changes and not changes in probabilities. The main result of this model is that being outsourced is related in average to approximately 0.5 points more in the index of good schedule. Considering that the index is originally delimited from 0 to 4, the change associated to outsourcing is of remarkable size. This estimate is statistically significant at the 99 % level. Considering that the good schedule index is composed of four dimensions, we can rerun the same analysis but now considering each dimension as outcome variable. The results from these estimations indicate that, considering sampling uncertainty, the relation between outsourcing and working at evenings or nights is virtually zero. The dependence we are observing in the third column of Table 3 comes from a positive and statistically significant association between being an outsourced worker and working during Saturdays or Sundays[3]. This is coherent considering that provider companies tend to adapt to the calendar and schedule of their client companies, being those mostly businesses developing their activities Monday to Friday and during social working hours. In contrast, in-house workers in ancillary occupations as cleaners may work during evenings and weekends as a large part of Spanish firms are part of the commerce and tourism sectors. As we can see, except the coefficient of the foreign level of *nationality*, the rest of coefficients are not statistically significant. For this variable, we find that compared with those with a European citizenship (other than the Spanish) foreign cleaners have in average better time-shifts.

Whereas the estimates of a multivariable model present coefficients of partial association, it is also interesting to consider the total association linked to each predictor. Table 4 presents the estimates of the Cramér's V for cleaners during year 2018. Regarding temporary employment, we see that the variable that collects more information to predict it is *age* ($\hat{V}_{Y_1 \sim A} = 0.108$). Our variable of interest, *outsourced*, is far from that estimate. Considering the previous results on partial associations, this points that the outsourcing status might be associated to bad predictors of temporary employment that we would need to held constant. Concerning part-time work, the predictive ability of the different variables is more similar. Consistent with the multivariable model, *age* and *sex* appear to be the best predictors. Still, their sizes are small and not so distant from the coefficient of outsourcing ($\hat{V}_{Y_2 \sim O} = 0.015$). Lastly, the estimates of the third column go

[3] The results from these additional estimations are available upon request.

in line with Table 3 and confirms that the best prediction of an outsourced situation is, in average, an improved schedule. The Cramér's V for outsourcing is the highest among the candidates, although of modest magnitude. Therefore, this indicates that outsourcing might be a good predictor of a better time-shifts after holding constant other covariates (partial association), but also on its own when considered alone (total association).

Table 4 – Estimates of Cramér's V for cleaners using LFS-2018

Covariate	*Temporary*	*Part-time*	*Good schedule*
Outsourced	0.011	0.015	0.104
Sex	< 0.001	0.021	0.003
Nationality	0.044	0.002	0.004
Age	0.108	0.026	0.006
Level of education	0.022	0.004	0.007
NUTS-1 region	0.01	0.004	0.015

Note: Own elaboration based on INE (2019). Variable *age* is categorized in six levels.

Secondly, we concentrate on the subsample of security guards and gardeners. Notice that the sample size did not allow us to control for the level of education or the NUTS-1 region. Table 5 shows the results of our estimations. Concerning temporary employment, as for cleaners, being outsourced is associated in average with a decrease in the probability of being temporary. This difference of probability between outsourced and in-house workers is of 11.5 p.p., being statistically significant at the 95 % level. We also found a statistically significant difference among the probabilities of being a temporary worker in function of being Spanish. The average marginal change is of 16.6 p. p. less, but we should be cautious as it is only significant at the 90% level. Variable *age* seems to add non-redundant information if we want to predict fixed-term contracts. To interpret its coefficient, consider that a contrast in its interquartile range (52 vs. 42 years old) is related in average to an 8 p. p. decrease in the probability of being temporary.

Regarding the association of outsourcing and part-time work, the group of guards and gardeners seem to have a different behaviour in comparison to what we found with cleaners. As we can see in the second column of Table 5, the coefficient of outsourcing is not statistically significant. This indicates that we cannot reject that in-house and outsourced workers had the same average probability of participating in a part-time job. As is well known in the literature, holding constant the rest of covariates, women have in average higher probability of part-time work. We also detect through a linear term a negative association between age and part-time employment, although this time the size of the coefficient is lower.

Table 5 – Model estimates for security guards and gardeners using LFS-2018

	Temporary	Part-time	Good schedule
Outsourced (Yes vs. No)	-0.115**	0.046	0.331**
	(0.051)	(0.031)	(0.129)
Sex (Woman vs. Man)	-0.026	0.169**	0.169
	(0.070)	(0.075)	(0.194)
Nationality (Spanish vs. not Spanish)	-0.166*	-0.037	-0.081
	(0.096)	(0.059)	(0.175)
Age	-0.008***	-0.003**	-0.002
	(0.002)	(0.001)	(0.006)
Num. Obs.	368	440	432
Design-based R^2	0.167	0.120	0.316

Note: Own elaboration based on INE (2019). Each cell reports the average marginal change with its standard error in parenthesis. The third model includes a coefficient for the occupation. For logistic regression models, the design-based version of the pseudo R^2 of Nagelkerke is shown. * p < 0.1, ** p < 0.05, *** p < 0.01.

The third model in Table 5 tried to estimate partial associations in order to predict the index of good schedule. The main finding of this model is that, for guards and gardeners, outsourcing is again associated with better schedules, but this time the coefficient is lower. Remind that this specification also controls for the occupation, given the different prevalences of night work intrinsic to security jobs. Specifically, being outsourced is in average related to a 0.33 increase in the score of good schedules, a difference that is statistically significant at the 95 % level. This result is in the same direction of what we found for cleaners. Still, it is true that the set of controls employed for both specifications are not the same, so comparisons are delicate. The rest of coefficients presented in the table are not statistically significant.

Lastly, Table 6 presents the results for the estimation of total associations of each predictor with each outcome of interest for the subsample of guards and gardeners. The findings are quite similar to what we detected for cleaners. Concerning temporary work, outsourcing again plays a marginal role as good predictor on its own. Other variables like age seem to collect more information without the requirement of being inserted in a multivariable model. Something similar occurs with the part-time outcome, although now all the candidate predictors have modest total associations with the response variable. Sex is highlighted again as the best predictor for part-time. Outsourcing really shines as the best predictor for the dimension of good schedule. The total association is not large ($\hat{V}_{Y_3 \sim O} = 0.071$), but also for these occupations serve as the variable with the highest relationship with the index of good schedule.

Table 6 – Estimates of Cramér's V for security guards and gardeners using LFS-2018

Covariate	*Temporary*	*Part-time*	*Good schedule*
Outsourced	0.019	0.007	0.071
Sex	0.001	0.04	0.016
Nationality	0.049	0.001	0.027
Age	0.11	0.015	0.023
Level of education	0.015	0.03	0.042
NUTS-1 region	0.053	0.009	0.031

Note: Own elaboration based on INE (2019). Variable *age* is categorized in six levels and *nationality* in three levels.

6. Conclusions

The observed data regarding time-related dimensions of job quality shows ambivalent results. Nevertheless, we can synthesize the analysis into three scenarios to identify different paradigmatic situations for outsourced workers compared to in-house employees. First of all, we found a scenario of better working conditions for outsourced workers due to institutional frameworks statutory mandatory. Coordination mechanisms can intervene in buffering market challenges and minimizing inequalities between workers of internal and outsourced services, such as the application of Transfer of Undertakings-Protection of Employment legislation, further impacting in lower levels of temporary work within outsourced workers covered by labour subrogation clauses in collective agreements.

Secondly, we found a scenario of better working conditions for outsourced workers in comparison to in-house employees due to the alignment of provider companies to certain practices of client organizations. This is the case of outsourced employees adjusting their working hours to the scheduling of these client companies. In this way, blue-collar outsourced workers engaged in auxiliary activities end up benefiting from this adaptation, in a flexible way, to standardized working hours for white-collar workers engaged in core activities of client companies.

And lastly, a common scenario of worse working conditions for outsourced workers in comparison to in-house employees is due to the implementation of labour cost-cutting strategies, particularly through the increasing of part-timers in these outsourced activities and the use of overtime work as adjustment mechanisms to the client's demand. Supplier companies operate based on economies of scale, optimizing work processes through high specialization and a concentration of clients with similar demands, thereby reducing costs for these non-core functions

for organizations. However, these supplier companies also optimize resources through a reduction in labour costs and the arbitrary application of collective agreements that allow for a discretionary implementation of minimum working hours. Precisely, the limited leeway for facility services provider companies to dismiss employees due to the mandatory application of labour subrogation clauses leads these companies to concentrate their strategies for adjusting labour costs on other employment dimensions. Primarily, this involves reducing working hours, thereby reducing salary costs, while commonly intensifying the work pace to maintain the contracted service by client companies.

The chapter also contains important policy insights. More specifically, ensuring the predominance of sectoral collective bargaining would help to set limits on wage-undercutting strategies. Another element contributing to reaching this goal is the implementation of national and EU regulations on tendering, public procurement, and codes of conduct. This is the case for the Barcelona City Council, which implements a social assessment of its providers and gives a weighted importance of 30% to social criteria and 70% to the offered price (Ajuntament de Barcelona, 2016). Thus, if the City Council detects that the offered price is abnormally low (and therefore implies working conditions below those agreed by collective negotiations), the offer will be excluded.

In that sense, it would be appropriate to consider these types of mechanisms across all government administrations. One way would be through the State's Advisory Board on Administrative Contracting. This body makes both recommendations and instructions to public administrations regarding the public contracting of third parties. One of the mechanisms of this board is precisely the establishment of a series of requirements for companies that are contracted (from a certain amount in construction and service contracts). These requirements are primarily financial, but we understand that, following the example of the Barcelona City Council, the introduction of new requirements for companies is necessary, mainly concerning the application of collective agreements, thereby conducting a prior social assessment.

These policy pointers can not only imply improvements in terms of job quality, but also in the quality of the service provided, particularly in the case of essential workers (as is the case of many facility services workers). Particularly in the health sector, Bacigalupe et al. (2015) found that the implementation of austerity and privatization policies during the great recession followed parallel patterns in Spain. This was later evidenced during the first wave of the pandemic in 2020, when there was a correlation between the incidence of COVID-19 mortality in residential care service and the trend to privatizing and increasing the number of users in nursing homes (Barrera-Algarín et al., 2021).

Beyond the results achieved, this study has several limitations that should be highlighted to address them in future research. In substantive terms, we envision two possible streams of future

research. Firstly, the study is limited to the Spanish case, which restricts the generalization of the results considering that FM practices vary significantly across countries. Therefore, further research should address comparisons with other countries with diverse stages of FM market development. Secondly, there are other time-related dimensions relevant to the improvement of job quality such as overtime, work intensity, schedule predictability, and control over working time. Although the consideration of the three variables analysed in the chapter responds to a focus in the literature precisely on stability, working time and unsocial schedule (particularly in the case of those theoretical approaches that have paid more attention to time-related dimensions), future steps expanding the current research should widen the dimensions considered.

In methodological terms, we think that there are two possible paths to continue the research. On the one hand, the objective of the analysis might move from description to effect estimation. To estimate the impact of outsourcing at firm level, Bilar and Lhuillier (2021) used an instrumental variable strategy to bypass the selection bias that presumably confounds conventional regressions. At the worker level, it is possible that jobseekers with low productivity self-select in a contractor firm because they tend to pay less and also accept worse job quality conditions for the same reason. On the other hand, this causal inference strategy might also be assisted by the analysis of qualitative data gathered through interviews or focus groups. The potential of these extensive datasets is that they can support or raise suspicions on the causal assumptions of an instrumental variable identification. Beyond evaluating the exogeneity assumption of instruments, qualitative data might also suggest hypotheses on the mechanisms (causal paths) that might explain the effects estimated with instrumental variable regressions.

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